

Derivatives of Trigonometric Functions

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Name: _____ Date: _____ Score: _____

Find the derivative of each function. Show all work.

Calculus 1 Worksheet #16 Derivatives of Trigonometric Functions

Notes: Know the following theorems.

1. $\frac{d(\tan \square)}{dx} = \sec^2 \square \cdot \frac{d\square}{dx}$	2. $\frac{d(\cot \square)}{dx} = -\csc^2 \square \cdot \frac{d\square}{dx}$	3. $\frac{d(\sec \square)}{dx} = \sec \square \cdot \tan \square \cdot \frac{d\square}{dx}$	4. $\frac{d(\csc \square)}{dx} = -\csc \square \cdot \cot \square \cdot \frac{d\square}{dx}$
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Examples:

1. $y = \tan 5x$ $y' = 5 \sec^2 5x$	2. $y = \sec 5x$ $y' = 5 \tan 5x \sec 5x$	3. $y = \cot^4 3x$ $y' = 4[-\cot^3 3x \csc^2 3x](3)$ $y' = -12 \cot^3 3x \csc^2 3x$	4. $y = \csc^3 2x$ $y' = 3(\csc^2 2x)[- \csc 2x \cot 2x](2)$ $y' = -6(\csc^2 2x) \csc 2x \cot 2x$
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Use the **quotient rule** to **prove** the derivative of:

1. $\tan x$ 2. $\cot x$ 3. $\sec x$ 4. $\csc x$

Directions: Find dy/dx.

5. $y = \sec 4x$	6. $y = \tan 3x - \cot 3x$	7. $y = \cot 5x + \csc 5x$
8. $y = \csc^3(2x)$	9. $y = \tan x + \cot x$	10. $y = 4 \sec x - 2 \csc x$
11. $y = 3 \sec x (\tan x)$	12. $y = \sin x (\tan x)$	13. $y = \cot x (\csc x)$
14. $y = \cos x (\cot x)$	15. $y = \frac{2 \cos x}{x+1}$	16. $y = \frac{\sin x}{x}$
17. $y = \frac{\sin x}{1 - \cos x}$	18. $y = \frac{x+2}{\cos x}$	19. $y = \frac{\tan x}{\cos x - 4}$
20. $y = \frac{\cot x}{1 - \sin x}$		



Derivatives of Trigonometric Functions

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Answer key — for instructor use only.

Answers:

1. $\sec^2 x$	2. $-\csc^2 x$	3. $\sec x \tan x$
4. $-\csc x \cot x$	5. $4 \sec 4x \tan 4x$	6. $3(\sec^2 3x + \csc^2 3x)$
7. $-5 \csc 5x (\csc 5x + \cot 5x)$	8. $-6 \csc^3(2x) \cot(2x)$	9. $\sec^2 x - \csc^2 x$
10. $2(2 \sec x \tan x + \csc x \cot x)$	11. $3 \sec x (\tan^2 x + \sec^2 x)$	12. $\sin x (1 + \sec^2 x)$
13. $-\csc x (\csc^2 x + \cot^2 x)$	14. $-\cos x (1 + \csc^2 x)$	15. $\frac{-2(x \sin x + \sin x + \cos x)}{(x+1)^2}$
16. $\frac{x \cos x - \sin x}{x^2}$	17. $\frac{1}{\cos x - 1}$	18. $\frac{\cos x + x \sin x + 2 \sin x}{\cos^2 x}$
19. $\frac{\sec x - 4 \sec^2 x + \tan x \sin x}{(\cos x - 4)^2}$	20. $\frac{-\csc^2 x + \csc x + \cot x \cos x}{(1 - \sin x)^2}$	

